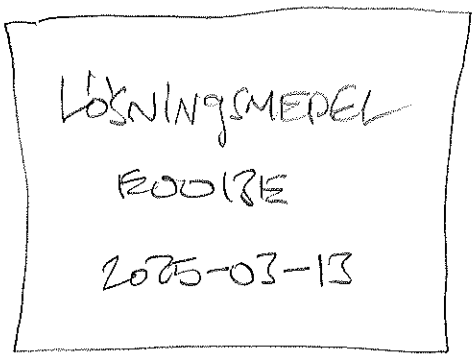
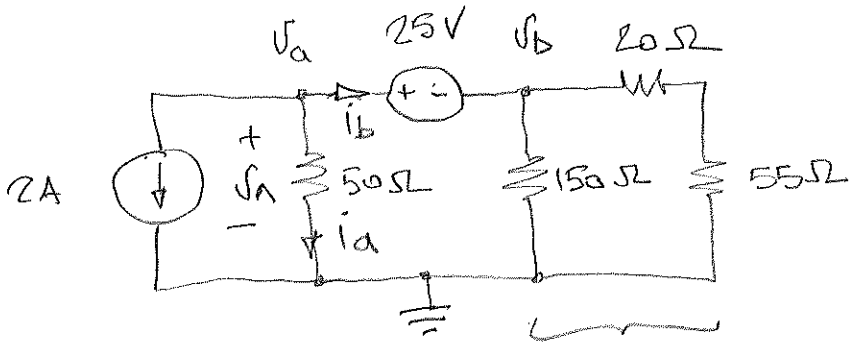
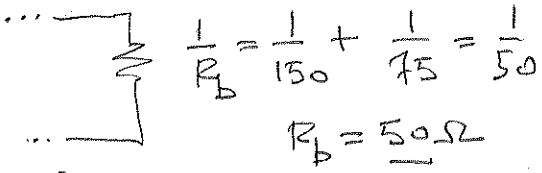

LÖSNINGSMEDEL
ROOIBE
2025-03-13

1.

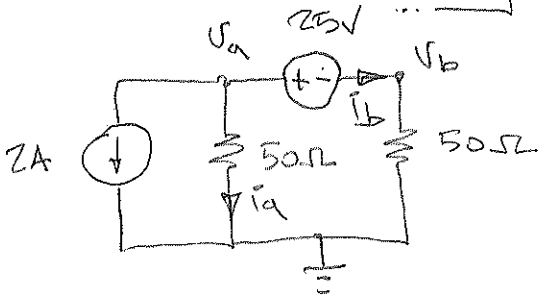


FÖRENKLA GENOM ATT SLÅ IHOP MOTSTÅNDEN OCH ATT SLÅ IHOP STRÖMMARNA



$$\frac{1}{R_b} = \frac{1}{150} + \frac{1}{75} = \frac{1}{50}$$

$$R_b = 50\,\Omega$$



KIRCHHOFFS STRÖMLAG I NOD a GER

$$-2 - i_a - i_b = 0 \qquad (1)$$

DÄR $i_a = \frac{v_a}{50}$, $i_b = \frac{v_b}{50}$ (2)

KSPL GER

$$50 i_b + 25 - 50 i_a = 0 \qquad (3)$$

(1) & (3)

$$50(-2 - i_a) + 25 - 50 i_a = 0$$

$$-100 - 50 i_a + 25 - 50 i_a = 0$$

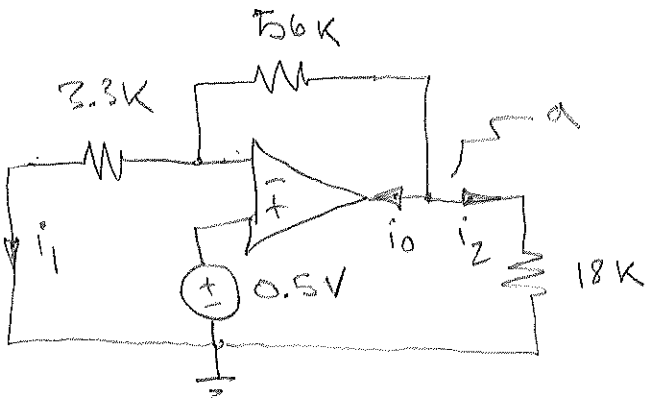
$$i_a = -\frac{75}{100}\ \text{A}$$

(2) MED (3) GER

$$v_a = 50\left(-\frac{75}{150}\right) = -\frac{75}{2} = -37.5\ \text{V}$$

SVAR: $v_a = -37.5\,\text{V}$

2.



KSTL INOD A

$-i_0 - i_1 - i_2 = 0$

VÄNSTRA SIDAN

$\frac{1}{2} - 3300 i_1 = 0$

$i_1 = \frac{1}{6600} \simeq 0.152$ mA

Högra sidan

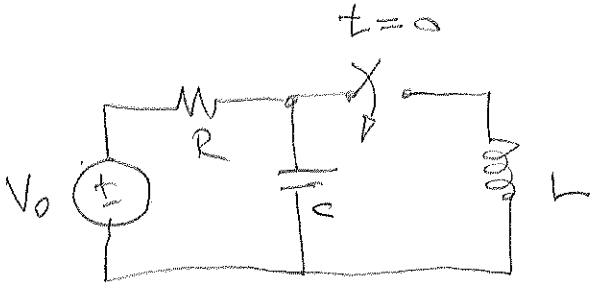
$\frac{1}{2} + 56000 i_1 - i_2 18000 = 0$

$i_2 = \frac{\frac{1}{2} + 56000 \cdot \frac{1}{6600}}{18000} \simeq 4.992 \cdot 10^{-4}$ A

slutligen $i_0 = -i_1 - i_2 = -\frac{1}{6600} - 4.992 \cdot 10^{-4} \simeq -6.507 \cdot 10^{-4}$ A

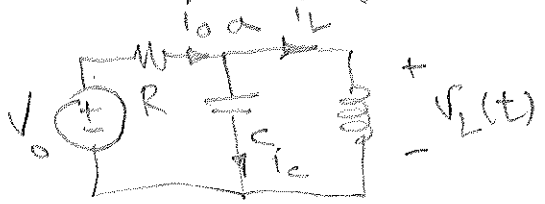
SVAR: $\begin{cases} i_0 \simeq -6.51 \cdot 10^{-4} A \\ i_1 \simeq 1.52 \cdot 10^{-4} A \\ i_2 \simeq 4.99 \cdot 10^{-4} A \end{cases}$

3.



För $t \leq 0$ gäller att $v_C = v_{C0} = V_0$ (1)

Då Brytaren sluts ($t \geq 0$)



För nod a gäller

$$i_0 - i_C - i_L = 0 \quad (2)$$

$$\begin{cases} i_L = i_L(0) + \frac{1}{L}\int v_L\,dt \\ i_C = C\frac{dv_C}{dt} = C\frac{dv_L}{dt} \\ i_0 = \frac{1}{R}(V_0 - v_a) = \\ \quad = \frac{1}{R}(V_0 - v_L) \end{cases} \quad (3)$$

(3) i (2) ger

$$\frac{1}{R}(V_0 - v_L) - C\frac{dv_L}{dt} - i_L(0) - \frac{1}{L}\int v_L\,dt = 0 \quad (4)$$

$i_L(0) = 0$ Ty spolen kan strömmen ändra momentant $i_L(t<0) = 0$

Deriverar (4) map t

$$\frac{1}{R}\frac{dv_L}{dt} + C\frac{d^2v_L}{dt^2} + \frac{1}{L}v_L = 0$$

Standard

$$\frac{d^2v_L}{dt^2} + \underbrace{\frac{1}{RC}}_{2\alpha}\frac{dv_L}{dt} + \underbrace{\frac{1}{LC}}_{\omega_0^2}v_L = 0$$

Siffror

$$\begin{cases} \alpha = \frac{1}{2RC} = \frac{1}{2\cdot 16\cdot 312.5\cdot 10^{-6}} \approx 100 \\ \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{2}\cdot 312.5\cdot 10^{-6}}} \approx 80 \end{cases}$$

$\alpha > \omega_0$ ÖVERDÄMPAD LÖSNING

$$v_L(t) = Ae^{-\xi_1 t} + Be^{-\xi_2 t}$$

DÄR

$$\xi_{1,2} = \alpha \pm \sqrt{\alpha^2 - \omega_0^2} \Rightarrow \begin{cases} \xi_1 = 40 \\ \xi_2 = 160 \end{cases}$$

$$\Rightarrow v_L(t) = Ae^{-40t} + Be^{-160t} \qquad (5)$$

KONSTANTERNA A & B ? SPÄNNINGEN ÖVER INDUKTANSEN KOMMER ATT VARA SAMMA SOM SPÄNNINGEN ÖVER KAPACITANSEN FÖRE $t > 0$

DVS

$$v_L(0) = v_C(0) \Rightarrow A + B = V_0 \qquad (6)$$

MED (4) FÅS FÖR $t = 0$

$$\frac{1}{R}(V_0 - v_L(0)) + C\left.\frac{dv_L}{dt}\right|_{t=0} = 0 \qquad (7)$$

(5) & (7) ger

$$\frac{1}{R}(V_0 - (A+B)) + C(-40A - 160B) = 0$$

$$\frac{1}{16}(4 - (A+B)) + C(-40A - 160B) = 0$$

$$\frac{1}{4} = \frac{1}{16}(A+B) + C(40A + 160B)$$

$$\frac{1}{4} = \frac{1}{16}A + \frac{1}{16}B + 40CA + 160CB \qquad (8)$$

ANVÄND (6)

$$\frac{1}{4} = \frac{1}{16}A + \frac{1}{16}(V_0 - A) + 40CA + 160C(V_0 - A) =$$

$$= \frac{1}{16}V_0 + 40CA + 160CV_0 - 160CA =$$

$$= \frac{1}{16}V_0 + 160CV_0 - 120CA$$

$$\Rightarrow \frac{1}{4} - \frac{1}{16}V_0 - 160CV_0 = -120CA$$

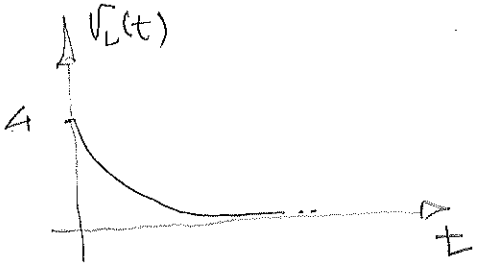
$$A = \frac{\frac{1}{4} - \frac{1}{16}V_0 - 160CV_0}{120C}$$

$$A = -\frac{\frac{1}{4} - \frac{1}{16}4 - 160 \cdot 312.5 \cdot 10^{-6} \cdot 4}{120 \cdot 312.5 \cdot 10^{-6}} \simeq +5.33$$

$$\Rightarrow B = V_0 - A \simeq 4 - 5.33 \simeq -1.33$$

ALTSÅ, MED (5) FÅS

$$v_L(t) = +5.33e^{-40t} - 1.33e^{-160t} \text{ volt.}$$



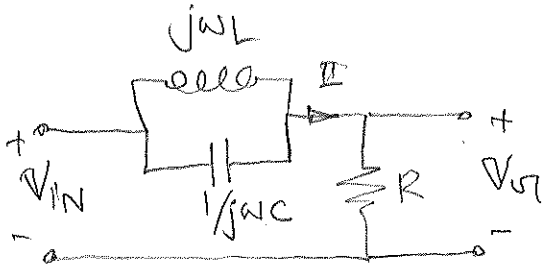
... ETT RIMLIGT ORIMLIGT FÖRLOPP.

SVAR:

$$v_L(t) = \frac{4}{3}(4e^{-40t} - e^{-160t}) \text{ V.}$$

FIGUR, SE OVAN.

4.



FÖRST P-KOPPLINGEN

$$\frac{1}{Z_p} = \frac{1}{j\omega L} + j\omega C = \frac{1 - \omega^2 LC}{j\omega L}$$

$$\Rightarrow Z_p = \frac{j\omega L}{1-\omega^2 LC} \qquad (0)$$

KSPL GER

$$\begin{cases} V_{IN} - Z_p II - R II = 0 & (1) \\ V_{UT} - R II = 0 \Rightarrow II = \frac{V_{UT}}{R} & (2) \end{cases}$$

(2) I (1)

$$V_{IN} - (Z_p + R)\frac{V_{UT}}{R} = 0$$

$$\frac{V_{UT}}{V_{IN}} = \frac{R}{Z_p + R} \qquad (3)$$

(3) MED (0)

$$\frac{V_{UT}}{V_{IN}} = H(\omega) = \frac{R}{\frac{j\omega L}{1-\omega^2 LC} + R} = \frac{R(1-\omega^2 LC)}{j\omega L + R(1-\omega^2 LC)} \qquad (4)$$

TAG BELOPP

$$|H(\omega)| = \frac{R(1-\omega^2 LC)}{\sqrt{(\omega L)^2 + R^2(1-\omega^2 L)}} \qquad (5)$$

$V_{UT}(\omega_c) = 0$ INTRÄFFAR DÅ TÄLJAREN $= 0$, DVS

$$1 - \omega_c^2 LC = 0$$

$$\omega_c = \frac{1}{\sqrt{LC}}$$

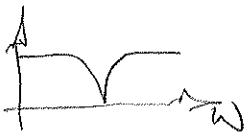
MED SIFFROR

$$\omega_c = \frac{1}{\sqrt{10 \cdot 10^{-3} \cdot 100 \cdot 10^{-9}}} \approx 31622.78 \text{ RAD/SEK}$$

SVARS:

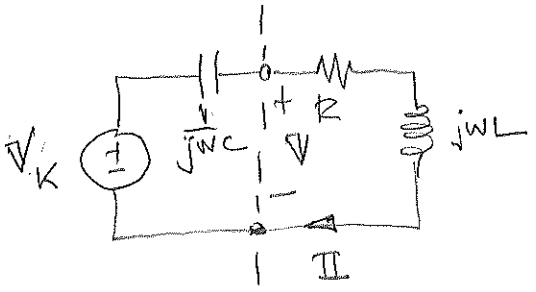
$$\begin{cases} H(\omega) = \dfrac{R(1-\omega^2 LC)}{j\omega L + R(1-\omega^2 LC)} \\ \omega_c \simeq 31.6 \text{ KRAD/SEK} \end{cases}$$

|H(ω)|



ETT BANDSPÄRRFILTER

5.



$v_K(t) = 150\cos(250t)$ V

$\begin{cases} C = 80\,\mu F \\ R = 50\,\Omega \\ L = 100\,mH \end{cases}$

STRÖMMEN II ges AV

$$II = \frac{V_K}{Z_{tot}} = \frac{V_K}{\frac{1}{j\omega C} + R + j\omega L} =$$

$$= \frac{150\angle 0^\circ}{-j50 + 50 + j25} = \frac{150\angle 0^\circ}{50 - j25} =$$

$$= \frac{150\angle 0^\circ}{55.90\angle -26.57^\circ} = \underline{2.683\angle +26.59^\circ}\ A.$$

SPÄNNINGEN ÖVER LASTEN BLIR

$$V_L = Z_L II = (R + j\omega L)II = (50 + j25)2.683\angle +26.57^\circ =$$

$$= 55.90\angle +26.57 \cdot 2.683\angle 26.57^\circ \simeq \underline{149.97\angle 53.14^\circ}\ V$$

KOMPLEX EFFEKT

$$\$_L = \frac{1}{2} V_L II^* = \frac{1}{2} 149.97\angle 53.14^\circ \cdot 2.683\angle -26.57^\circ =$$

$$= \underline{201.18\angle 26.57^\circ}\ VA$$

SVAR: $\$_L = 201.2\angle 26.57^\circ$ VA